Plays and Payoffs in Dialectica Games*

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Abstract

A dialectica object is a set of witnesses, a set of challenges and a relation saying which witness withstands which challenge. Read as a game, the witnesses are the strategies of one player, the challenges are the strategies of the other, and the relation is the payoff: a game in normal form, where each player commits once and nothing is seen before it is chosen. The objection to this reading is that games are about taking turns (or alternation of players), and normal form has none. We answer the objection from inside the construction. The substitution tensor is one round of question and response, its iterates are bounded alternation, and the interchange law says that independent dialogues can be scheduled into common rounds. The modality for sequential reuse is a commitment that must survive a whole history of challenges, and the cofree comonoid on a dialectica object has for its witnesses the responses to every finite history, strategies that depend on the history of play. So the passage from normal-form strategies to history-dependent ones can be made inside the category, rather than only across an adjunction out of it. The payoff takes its values in a lineale and need not be a win or a loss. The value of a game, the payoff the proponent can guarantee, is then a lax monoidal functor to the lineale. When the multiplication of the lineale preserves meets, as it does over the booleans, a Heyting algebra or the Lawvere quantale, two games played side by side and the same two games played in alternation have the same value; over other lineales alternation can raise it, and a powerset lineale gives an example.

*This note starts from the work of the games working group of the 2022 Mathematics Research Community on Applied Category Theory, Colin Bloomfield, Jérémie Koenig, Jan Rooduijn and the author: the assumption that Dialectica is to game semantics as normal-form games are to extensive-form games, and the programme of comparing dialectica objects with a category of extensive-form games. That work was presented by Jérémie Koenig at the Joint Mathematics Meetings in January 2023 [9].

1 Introduction

A dialectica object has the two players of a game: the witnesses are the strategies of a proponent, the challenges those of an opponent, and the relation says which strategy wins against which. That is a game in normal form, but it leaves out the plays, the moves of the game, in which each player can respond to what the other has done. This note asks how much of the play can be recovered inside the Dialectica construction, and what difference it makes to the payoff. The answer has two parts, according to which player is given information: a witness may depend on earlier challenges, or a challenge may depend on the witness.

Games entered mathematics as rules for moves. Zermelo's paper on chess [17] treats a game as a succession of positions and asks what a player can force from a given one. Von Neumann's paper of 1928 [15] describes a game of strategy by its rules: a sequence of moves, each made by one of the players or by chance, with each player informed of some part of what has happened before. His decisive step was to set that description aside. A strategy is a complete plan for every situation the rules allow, so each player may be taken to choose one strategy, once, without seeing the choice of the other, and the game becomes a function from pairs of strategies to payoffs. The minimax theorem is proved for that function.

Von Neumann and Morgenstern made the two descriptions explicit [16], calling them the extensive and the normalized form, and built their theory on the second. The passage between them goes one way. Every game in extensive form has a normal form, obtained by turning each player's moves into a choice of plan, but a payoff matrix does not say which game it came from or in what order its choices were made. Kuhn [10] replaced the set-theoretic description by a tree whose nodes are grouped into information sets, and showed that when players remember what they have done and seen, nothing is lost by letting them choose at each move rather than once at the start. Later game theory needed the tree back: Selten's refinements of Nash equilibrium [14] exclude threats a player would not carry out once the play had reached them, and that cannot be said without the order of moves.

The Dialectica reading of a proposition [6] asks for a witness u such that $\alpha(u, x)$ holds for every challenge x: a choice made once, against every possible reply, which is a payoff matrix searched for a winning row. This reading has been part of the construction from the start. When the construction was first presented, Girard remarked (personal communication) that his variant corresponds to Henkin's branching quantifiers [8]. The in-

ternal hom of $\mathsf{Gir}(\mathbf{Set})$ asks for $f\colon U \to V$ and $F\colon Y \to X$, so each player's choice depends only on the other player's choice in its own branch. That is the prefix

$$\begin{pmatrix} \forall u\ \exists v \\ \forall y\ \exists x \end{pmatrix}$$

of a branching quantifier, and a game in which neither player sees the other. In $\mathbf{Dial}(\mathbf{Set})$ the backward map is $F\colon U \times Y \to X$, the prefix becomes the linear $\forall u\, \exists v\, \forall y\, \exists x$, and one alternation is present before any connective is applied. Blass wrote the game reading down for his category of questions and answers [2], and Hedges read a dialectica object over a category of games as a round of bidding that selects which game is then played [7], working in the Girard variant. This work's assumption, as presented by Koenig in 2023 [9], is that Dialectica is to game semantics as normal-form games are to extensive-form games. This note works in $\mathbf{Dial}_L(\mathbf{Set})$.

The substitution tensor $\triangleleft$ is one round of question and response, and its iterates are alternation of bounded depth (Section 3). Unbounded alternation is the object HA whose witnesses respond to every finite history of challenges, and HA is the cofree $\triangleleft$-comonoid on A (Section 4). The map from the modality $\dagger A$, whose witness is a single commitment kept throughout a history, into HA is then forced by the universal property: the comparison between normal-form and extensive-form strategies is part of the structure rather than a translation out of it. The object HA does not recover Kuhn's general tree. The challenges of a dialectica object form one set, so every round offers the same moves, and the proponent sees the whole history. Games whose available moves depend on the position reached, or in which a player sees only part of what has happened, ask for challenges indexed by witnesses, which is a change to the objects rather than to the connectives.

The predicate of a dialectica object takes values in a lineale [5], and the value of a game, the payoff the proponent can guarantee, is a lax monoidal functor to that lineale (Section 5). Over the booleans, over a Heyting algebra and over the Lawvere quantale, seeing the first challenge buys the proponent nothing in value: $A \otimes B$ and $A \triangleleft B$ differ in their strategies and not in what those strategies guarantee. Over lineales whose multiplication does not preserve meets the two values can differ, and a powerset lineale gives an example. A game with a payoff for each player is a pair of predicates on the same witnesses and challenges, transported by the same morphisms; that is an object over the product lineale, the setting in which Elena Di Lavore, Wilmer Leal and de Paiva modelled Petri nets, whose category of nets is $\mathsf{Gir}_{L\times L}(\mathbf{Set})$ [11]. Saddle points, the equilibria of games with a single

payoff, are closed under $\otimes$ and $\triangleleft$.

2 Normal forms

A dialectica object is a triple $A = (U, X, \alpha)$ with U and X sets and α a relation between them [3]. Read as a game, a member of U is a strategy for the proponent, a member of X is a counter-strategy for the opponent, and α evaluates the pair: the object is a payoff matrix. Truth of A is the existence of a strategy that wins against every counter-strategy, that is $\exists u \in U.\ \forall x \in X.\ \alpha(u, x)$, which is the Dialectica reading of a proposition and the game-theoretic reading of a winning strategy at once. This is the normal-form reading of [9].

The connectives read as the usual operations on games. In $A \,\&\, B$ the proponent must be prepared for either game and the opponent picks which one to play, so the witnesses are pairs and the challenges are a disjoint union: a challenge names a component and challenges it. In $A \otimes B$ the two games are played side by side with nothing passing between them, so both witnesses and challenges are pairs and the payoff is the product of the two payoffs. In $!A$ the opponent may instantiate several copies of the same game against a single commitment of the proponent, so a challenge is a finite multiset of challenges to A and the payoff is the product over its entries; the copies are unordered because they are played in parallel. Swapping the roles of the two players is the linear negation, which is available in the Girard construction $\mathsf{Gir}(\mathbf{Set})$ rather than in $\mathbf{Dial}(\mathbf{Set})$, and it takes the tensor to the par.

A small example runs through the note. Let a payoff be a set of possible scores, combined by adding scores, so that the payoff of two rounds is $\{s+t\}$ for the scores s and t of the rounds. In the game $A = (1, \{x_1, x_2\}, \alpha)$ the proponent has nothing to choose and the opponent's two challenges score $\alpha(x_1) = \{0\}$ and $\alpha(x_2) = \{1\}$. In the game $B = (\{v_1, v_2\}, 1, \beta)$ the opponent has nothing to choose and the proponent's two witnesses score $\beta(v_1) = \{1\}$ and $\beta(v_2) = \{0\}$. Played side by side, as $A \otimes B$, the proponent commits to v_1 or v_2 before the challenge is known, and either commitment leaves two possible totals. Played in alternation, as $A \triangleleft B$ in Section 3, the proponent answers x_1 with v_1 and x_2 with v_2, and the total is always 1.

A morphism from A to B is a way of playing B by playing A: a map f turning a proponent strategy for A into one for B, and a map F turning an opponent strategy for B, in the light of the proponent strategy being used, into one for A, such that winning in A against the translated challenge implies winning in B against the original. That the backward map may see

the witness is what makes the reading below possible. The functor from $\mathbf{Dial}(\mathbf{Set})$ to $\mathsf{Gir}(\mathbf{Set})$ sending (U, X, α) to $(U, X^U, \hat{\alpha})$, with $\hat{\alpha}(u, f)$ holding when $\alpha(u, fu)$ does, is left adjoint to the functor from $\mathsf{Gir}(\mathbf{Set})$ to $\mathbf{Dial}(\mathbf{Set})$ that is the identity on objects, since a morphism of $\mathsf{Gir}(\mathbf{Set})$ out of $(U, X^U, \hat{\alpha})$ is a morphism of $\mathbf{Dial}(\mathbf{Set})$ out of (U, X, α) [3]. The comonad it induces on $\mathsf{Gir}(\mathbf{Set})$ is the reader comonad of the Dialectica literature, the factor that appears in the modality of $\mathsf{Gir}(\mathbf{Set})$ [4], and its co-Kleisli category is $\mathbf{Dial}(\mathbf{Set})$.

The objection to all of this is that games are about alternation and a payoff matrix has none beyond the single step just described. One answer is external: build a category of games from the rules 1, $\bot$, a choice by the proponent and a choice by the opponent, send a dialectica object to the game in which the proponent chooses u, the opponent answers x and the payoff decides, flatten a game back to the pair of its strategy sets with the evaluation map, and ask for an adjunction. That is the programme of the talk, and the interesting part of it is where the two directions fail to be monoidal. The rest of this note gives an internal answer, using structure the construction already has, and Section 3 ends with the interchange map, which is a lax comparison of exactly the kind that programme asks for.

3 Alternation

This section adds alternation to dialectica objects by a second tensor, the substitution tensor, in which the witness of the second component may be chosen in the light of the challenge to the first. It is a tensor on $\mathbf{Dial}_L(\mathbf{Set})$ that is not commutative, and it interprets the ordered contexts of the linear logic model of state [4], as shown in [13]. That note develops it and compares it with the composition of polynomials, which has the same witnesses and challenges and no predicate; what follows recalls what is needed here.

The substitution tensor is one alternation. For dialectica objects $A = (U, X, \alpha)$ and $B = (V, Y, \beta)$, the substitution tensor is

$$A \triangleleft B \;=\; \big(\, U \times V^X,\; X \times Y,\; \alpha(u, x) \otimes \beta(\varphi x, y) \,\big),$$

so a witness is a witness u of A together with a rule φ assigning a witness of B to each challenge to A, and a challenge is a pair [13]. Read as a play, that is four moves. The proponent commits to u; the opponent answers with x; the proponent commits to φx, chosen in the light of x; the opponent answers with y. The payoff asks the proponent to win both rounds; the order of the rounds is carried by the witnesses, since φ sees x, and not by the lineale,

which is commutative. So the passage from $\otimes$ to $\triangleleft$ is the passage from two simultaneous commitments to one round of question and response.

Proposition 1 ([13])**.** *The substitution tensor is a bifunctor on* $\mathbf{Dial}_L(\mathbf{Set})$*, associative with unit* $I = (1, 1, e)$*, and not commutative. For* $(f, F)\colon A \to A'$ *and* $(g, G)\colon B \to B'$*, the morphism* $(f, F) \triangleleft (g, G)$ *is*

$$(u, \varphi) \mapsto \big(fu,\ x' \mapsto g\varphi F(u, x')\big), \quad (x', y') \mapsto \big(F(u, x'),\ G(\varphi F(u, x'), y')\big)$$

on witnesses and, at the witness (u, φ)*, on challenges.*

Its morphism condition follows from those of (f, F) and (g, G) and the monotonicity of $\otimes$: writing $v = \varphi F(u, x')$, it is

$$\alpha\big(u, F(u, x')\big) \otimes \beta\big(v, G(v, y')\big) \ \sqsubseteq\ \alpha'(fu, x') \otimes \beta'(gv, y').$$

Associativity holds on witnesses because $U \times (V \times W^Y)^X \cong (U \times V^X) \times W^{X \times Y}$ by currying, on challenges because the product is associative, and on predicates because $\otimes$ is; the unit laws hold because e is the unit of the lineale.

The map $\mathsf{ser}\colon A \otimes B \to A \triangleleft B$, which sends the witness (u, v) to (u, const_v) and leaves challenges alone, says that a simultaneous pair is an alternating pair whose second commitment ignores what the opponent did: a strategy that does not listen. It runs one way only. A strategy that listens cannot in general be replayed as one that does not. In the example of Section 2, the strategy of $A \triangleleft B$ that answers x_1 with v_1 and x_2 with v_2 guarantees the total 1, and neither of the two strategies that ignore the challenge guarantees any total.

Iterating gives bounded alternation. A witness of $A \triangleleft (B \triangleleft C)$ is a witness of A, a rule choosing a witness of B from a challenge to A, and a rule choosing a witness of C from the challenges raised so far, so the witnesses are $U \times (V \times W^Y)^X$ and the play has six moves. Associativity of $\triangleleft$ says that the bracketing does not matter: the same strategies are described whether one reads the object as a first game followed by a two-round game, or as a two-round game followed by a last response. Alternation of depth n is an n-fold nesting, and the challenges remain a plain product throughout, one challenge to each round.

Interchange schedules independent dialogues into rounds. The interchange map

$$\mathsf{int}\colon (A \triangleleft B) \otimes (C \triangleleft D) \longrightarrow (A \otimes C) \triangleleft (B \otimes D)$$

takes two alternating plays running side by side, each with two rounds of its own, to a single alternating play whose first round is the two opening moves together and whose second round is the two responses. Nothing passes between the two dialogues: the witness of B still depends only on the challenge to A, and the witness of D only on the challenge to C. The map imposes a common schedule. It runs one way only, and the direction is the informative one: independent plays can always be put into rounds, and plays already sharing rounds cannot be freed from them.

In the free setting the inclusions between series-parallel orders are generated by the middle interchange rule [1], so int is the comparison to expect between the two tensors; ser is its instance with B and C the unit. In game terms, the ways of imposing order on a collection of independent plays are generated by the one that synchronises two pairs into two rounds.

4 Strategies over histories

The modality for sequential reuse takes a dialectica object to

$$\dagger A \;=\; \big(U,\; X^*,\; \bigotimes_i \alpha(u, x_i)\,\big),$$

where X^* is the free monoid on the challenges. Read as a game, the proponent commits once and the opponent plays a whole sequence of challenges against that single commitment; the payoff is the product of the payoffs of the rounds. The challenges form a sequence, where those of $!A$ form a multiset, and the order of the sequence is what HA below responds to, one challenge at a time. The witness does not change as the history grows, which is the sense in which $\dagger A$ is storage that survives repeated use. In the $\triangleleft$-comonoid structure $\dagger A$ carries, the comultiplication sends u to (u, const_u), so the witness reached after any challenge is the one committed to.

A strategy that does change with the history can be written as a dialectica object too. Take

$$HA \;=\; \Big(U^{X^*},\; X^*,\; \gamma\Big), \qquad \gamma(\sigma, x_1 \cdots x_n) \;=\; \bigotimes_{i=1}^{n} \alpha\big(\sigma(x_1 \cdots x_{i-1}),\, x_i\big),$$

whose witnesses assign a commitment to every finite history of challenges, whose challenges are those histories, and whose payoff evaluates each round against the commitment the strategy makes in the light of what has been played before it; the empty history has payoff e, the unit of the lineale. That is an extensive-form strategy for the proponent, written as a dialectica

object. Sending u to the constant strategy gives a map $\dagger A \to HA$, since $\gamma(\mathrm{const}_u, w)$ is the product of $\alpha(u, x_i)$ over the letters of w, which is the payoff of $\dagger A$; so $\dagger A$ is the part of HA where the proponent has nothing to learn from being challenged.

Proposition 2. *The object HA is the cofree $\triangleleft$-comonoid on A in $\mathbf{Dial}_L(\mathbf{Set})$. Its counit sends every strategy to the empty history, its comultiplication sends σ to $(\sigma,\, w \mapsto \sigma_w)$ with $\sigma_w(v) = \sigma(wv)$ and pulls (w, v) back to wv, and the map $HA \to A$ sends σ to $\sigma(\varepsilon)$ and pulls x back to the one-letter history.*

Proof. The comonoid laws hold on witnesses and challenges because concatenation of histories is associative with the empty history as unit, and $(\sigma_w)_{w'} = \sigma_{ww'}$. For the comultiplication the predicate condition is $\gamma(\sigma, wv) \sqsubseteq \gamma(\sigma, w) \otimes \gamma(\sigma_w, v)$, which holds with equality: the rounds of v are evaluated at $\sigma(wv_1 \cdots v_{j-1}) = \sigma_w(v_1 \cdots v_{j-1})$. The counit and the map to A also hold with equality.

Let $C = (W, Z, \rho)$ be a $\triangleleft$-comonoid, with comultiplication $\delta \colon C \to C \triangleleft C$ and counit $\kappa \colon C \to I$, where $I = (1, 1, e)$ is the unit of $\triangleleft$ and e the unit of the lineale. The witnesses of $C \triangleleft C$ are pairs in $W \times W^Z$, and the counit law makes the first component of the forward part of δ the identity, so δ sends w to $(w,\, z \mapsto w \cdot z)$, where $w \cdot z$ is the witness reached from w after the challenge z. The backward part of δ sends a pair of challenges (z, z') at w to a single challenge $z; z'$, and the backward part of κ picks a challenge ι_w at each w. That δ and κ are morphisms says

$$\rho(w, z; z') \;\sqsubseteq\; \rho(w, z) \otimes \rho(w \cdot z, z') \qquad \text{and} \qquad \rho(w, \iota_w) \;\sqsubseteq\; e,$$

and the counit and coassociativity laws say that $w \cdot \iota_w = w$, that $\iota_w; z = z = z; \iota_{w \cdot z}$, that $(w \cdot z) \cdot z' = w \cdot (z; z')$ and that $(z; z'); z'' = z; (z'; z'')$, wherever these are defined. Given $(f, H) \colon C \to A$, put $w_0 = w$, $z_i = H(w_{i-1}, x_i)$ and $w_i = w_{i-1} \cdot z_i$. The induced map sends w to the strategy $\sigma^w \colon x_1 \cdots x_n \mapsto f(w_n)$, and pulls a history back by the recursion $k(w, \varepsilon) = \iota_w$ and $k(w, x_1 v) = z_1; k(w_1, v)$, which assembles the challenges z_i with the backward part of δ. Its predicate condition is

$$\rho\big(w, k(w, x_1 \cdots x_n)\big) \;\sqsubseteq\; \bigotimes_{i=1}^{n} \alpha(f w_{i-1}, x_i).$$

By induction on n the two inequalities above give

$$\rho\big(w, k(w, x_1 \cdots x_n)\big) \;\sqsubseteq\; \bigotimes_{i=1}^{n} \rho(w_{i-1}, z_i),$$

the empty case being $\rho(w, \iota_w) \sqsubseteq e$, and each factor is below $\alpha(fw_{i-1}, x_i)$ by the morphism condition for (f, H).

The induced map is a morphism of comonoids. On witnesses, comultiplying σ^w gives $(\sigma^w, v \mapsto (\sigma^w)_v)$, and comultiplying w in C before applying the map gives $(\sigma^w, v \mapsto \sigma^{w \cdot k(w,v)})$. These agree, because by the coassociativity and counit laws $w \cdot k(w, v)$ is the witness reached from w along v, so that $\sigma^w(vv') = \sigma^{w \cdot k(w,v)}(v')$. On challenges both sides pull (v, v') back to $k(w, vv') = k(w, v); k(w \cdot k(w, v), v')$, by induction on the length of v and associativity of ;. The counit is preserved because $k(w, \varepsilon) = \iota_w$, and composing with $HA \to A$ returns (f, H), because $\sigma^w(\varepsilon) = f(w)$ and $k(w, x) = z_1; \iota_{w_1} = H(w, x)$.

It is the only such morphism. Let (m', k') be a comonoid morphism $C \to HA$ whose composite with $HA \to A$ is (f, H), and put $\tau = m'(w)$. The composite gives $\tau(\varepsilon) = f(w)$ and $k'(w, x) = H(w, x)$ on a one-letter history. Preservation of the comultiplication gives $\tau_v = m'(w \cdot k'(w, v))$ on witnesses and $k'(w, vv') = k'(w, v); k'(w \cdot k'(w, v), v')$ on challenges, and preservation of the counit gives $k'(w, \varepsilon) = \iota_w$. Taking v to be a single letter, induction on the length of histories gives $\tau = \sigma^w$ and $k' = k$. □

On shapes, forgetting the predicate, HA is the cofree comonoid on Uy^X in the category of polynomial functors, which has the same witnesses and challenges [13]. The proof uses the witness in both backward maps, so it is a result about $\mathbf{Dial}_L(\mathbf{Set})$ and has no counterpart in $\mathsf{Gir}_L(\mathbf{Set})$. Since $\dagger A$ is a $\triangleleft$-comonoid over A, it has a unique comonoid map into HA; the witness reached from u after any challenge is u itself, so this is the constant-strategy map above.

So the three stages are one object each. Normal form is A, a single commitment against a single challenge. Bounded alternation is an iterate of $\triangleleft$, a commitment revised a fixed number of times. Unbounded alternation is HA, a commitment revised at every point of every history. The adjunction relates the first of these to a category of extensive-form games from outside; the second and third say how much of the passage can be made without leaving $\mathbf{Dial}(\mathbf{Set})$.

5 Payoffs

Nothing above requires the payoff to be a win or a loss. The predicate of a dialectica object may take its values in a lineale, a partially ordered commutative monoid with a residuation [5], and everything in the previous two sections was stated in those terms: the payoff of a play in $A \otimes B$ or in

$A \triangleleft B$ is the monoid product of the payoffs of its parts, and the order of the lineale is what the morphism condition compares. Taking the lineale to be the booleans returns the win-or-lose reading.

Two other choices are worth naming. Over the natural numbers extended with ∞, ordered by $\geq$ and with addition as multiplication, the payoff of a play is the sum of the costs of its rounds, so a strategy is graded by a cost and a morphism is a translation that does not increase it. The reversed order makes addition residuated, with truncated subtraction as residual, and ∞ makes the lineale complete. It is the integer part of the Lawvere quantale, so the results below apply to it, and there the value defined next is $\min_u \max_x \alpha(u, x)$, the least cost the proponent can guarantee. Over a three-element (non-commutative) lineale the order of the rounds is visible in the payoff itself, which is how a non-commutative connective is obtained without changing the shape of the object [4].

For a complete lineale L the value of a game $A = (U, X, \alpha)$ is $v(A) = \bigvee_u \bigwedge_x \alpha(u, x)$. Each witness u guarantees $\bigwedge_x \alpha(u, x)$, the worst payoff the opponent can force against it, and $v(A)$ is the join of these guarantees; when the join is attained, as over a chain with U finite, it is the best payoff the proponent can guarantee. A two-by-two game shows the calculation. Take payoffs in the chain $0 < 1 < 2 < 3$, higher being better for the proponent, and

α	x_1	x_2
u_1	3	1
u_2	2	2

Against u_1 the opponent answers x_2, so u_1 guarantees 1; against u_2 every answer gives 2, so u_2 guarantees 2, and $v(A) = 2$. The calculation uses only the order of L. Over 2 it says that $v(A) = 1$ exactly when $\exists u \, \forall x \, \alpha(u, x)$, so the value extends the Dialectica reading of truth to payoffs.

Proposition 3. *The value is a lax monoidal functor* $v \colon \mathbf{Dial}_L(\mathbf{Set}) \to L$, *and*

$$v(A) \otimes v(B) \;\sqsubseteq\; v(A \otimes B) \;\sqsubseteq\; v(A \triangleleft B).$$

When $\otimes$ *preserves nonempty meets and challenge sets are nonempty, all three are equal.*

Proof. A morphism $(f, F) \colon A \to B$ gives $\bigwedge_x \alpha(u, x) \sqsubseteq \bigwedge_y \alpha(u, F(u, y)) \sqsubseteq \bigwedge_y \beta(fu, y)$ for each u, hence $v(A) \sqsubseteq v(B)$. The first inequality uses only that $\otimes$ is monotone and preserves joins, and the second is v applied to

$\mathsf{ser}\colon A \otimes B \to A \triangleleft B$. If $\otimes$ preserves nonempty meets, then for each u and φ

$$\begin{aligned}\bigwedge_{x,y} \alpha(u,x) \otimes \beta(\varphi x, y) &= \bigwedge_x \Big(\alpha(u,x) \otimes \bigwedge_y \beta(\varphi x, y)\Big) \\ &\sqsubseteq \bigwedge_x \big(\alpha(u,x) \otimes v(B)\big) = \Big(\bigwedge_x \alpha(u,x)\Big) \otimes v(B),\end{aligned}$$

and taking the join over u and φ gives $v(A \triangleleft B) \sqsubseteq v(A) \otimes v(B)$. □

So over 2, over a Heyting algebra with $\otimes = \wedge$, and over the Lawvere quantale, seeing the first challenge buys the proponent nothing in value: $A \otimes B$ and $A \triangleleft B$ differ in their strategies and not in what those strategies guarantee. Over lineales where $\otimes$ does not preserve meets the second inequality can be strict, and the example of Section 2 is such a case. Its payoffs live in $\mathcal{P}(\mathbb{N})$ with complex sum, ordered by inclusion, whose multiplication does not preserve meets: $\{0,1\} + (\{0\} \cap \{1\}) = \emptyset$, while $(\{0,1\} + \{0\}) \cap (\{0,1\} + \{1\}) = \{1\}$. Every witness of $A \otimes B$ leaves two different singleton totals, so $v(A \otimes B) = \emptyset$, while the strategy answering x_1 with v_1 and x_2 with v_2 guarantees the total 1, so $v(A \triangleleft B) = \{1\}$.

Equilibria of a single payoff are saddle points. A dialectica object has one payoff, so as a game it is zero-sum, and an equilibrium is a pair (u^*, x^*) with

$$\alpha(u, x^*) \sqsubseteq \alpha(u^*, x^*) \sqsubseteq \alpha(u^*, x) \qquad \text{for all } u \in U,\ x \in X.$$

Such a pair exists exactly when $\bigvee_u \bigwedge_x \alpha$ and $\bigwedge_x \bigvee_u \alpha$ agree and are attained. In the two-by-two game above the opponent's side gives 3 against x_1 and 2 against x_2, so the upper value is 2, equal to $v(A)$, and (u_2, x_2) is a saddle point. Over 2, matching pennies, with $U = X = \{H, T\}$ and $\alpha(u,x)$ holding when $u = x$, has lower value 0 and upper value 1, and so no saddle point. Saddle points are closed under both tensors. If (u^*, x^*) and (v^*, y^*) are saddle points of A and B, then $((u^*, v^*), (x^*, y^*))$ is one of $A \otimes B$ and $((u^*, \mathrm{const}_{v^*}), (x^*, y^*))$ is one of $A \triangleleft B$, both by monotonicity of $\otimes$ in L. They are a property of objects and not something morphisms transport, since a morphism carries the proponent's choice forward and the opponent's backward. At a saddle point the payoff is the value, so when A and B both have saddle points the payoffs above give $v(A) \otimes v(B) = v(A \otimes B) = v(A \triangleleft B)$ over any lineale; in the example over $\mathcal{P}(\mathbb{N})$ the object A has none, since no single challenge attains its upper value $\emptyset$. A saddle point is a pure-strategy Nash equilibrium of the game in which the proponent maximises and the opponent minimises the single payoff. Games with general payoffs and their

equilibria are considered in the open games of Ghani, Hedges, Winschel and Zahn [12], which are morphisms of a symmetric monoidal category composed sequentially and in parallel, and whose equilibria are pure-strategy Nash equilibria when the players maximise their payoffs.

The substitution tensor gives information to the proponent, a witness chosen in the light of an earlier challenge. Information can be given to the opponent instead. In the cross product $A \otimes_{\mathsf{Gir}} B = (U \times V,\ X^V \times Y^U,\ \alpha(u, gv) \otimes \beta(v, hu))$ a challenge depends on a witness, and the value does not see it: for a fixed witness (u, v) the pairs (gv, hu) range over all of $X \times Y$, so $v(A \otimes_{\mathsf{Gir}} B) = v(A \otimes B)$ over any lineale. The extra information lowers only the upper value $\bigwedge_x \bigvee_u \alpha(u, x)$, the most the opponent can hold the payoff down to. The extreme case is the functor of Section 2, $R(A) = (U, X^U, \hat{\alpha})$, in which every challenge is chosen in the light of the whole commitment.

Proposition 4. *Let L be a complete lineale. The lower and upper values of $R(A)$ agree for every dialectica object A with U and X nonempty exactly when L is completely distributive, and the common value is then $v(A)$. When both are attained, in particular when L is a chain and U and X are finite, $R(A)$ has a saddle point.*

Proof. For a fixed u, as f ranges over X^U the challenge fu ranges over all of X, so $\bigwedge_f \alpha(u, fu) = \bigwedge_x \alpha(u, x)$ and the lower value of $R(A)$ is $v(A)$. The upper value is $\bigwedge_f \bigvee_u \alpha(u, fu)$, and the equation

$$\bigvee_u \bigwedge_x \alpha(u, x) \;=\; \bigwedge_{f\colon U \to X} \bigvee_u \alpha(u, fu)$$

for all families indexed by $U \times X$ is complete distributivity of L. A family whose index sets vary with u reduces to one indexed by a single set by padding with $\top$, which changes neither side. The last claim is the characterisation of saddle points above. □

Over 2 the proposition says that a game in which the opponent replies to the commitment is determined, which for games of two moves is Zermelo's observation [17]. Together with Proposition 3 it separates the two sides of a dialectica object. An exponent on the side of the witnesses can raise the value, and only over lineales whose multiplication does not preserve meets; an exponent on the side of the challenges leaves the value alone and closes the gap above it.

Two payoffs are one payoff in the product lineale. A game with a payoff for each player is a pair of predicates $\alpha_1, \alpha_2 \colon U \times X \to L$ on the same

witnesses and challenges, and asking one morphism to respect both is asking it to respect $\langle \alpha_1, \alpha_2 \rangle \colon U \times X \to L \times L$, since the product of lineales is a lineale with the componentwise structure. This is the setting of the Petri nets of Elena Di Lavore, Wilmer Leal and de Paiva [11], where the two predicates are the pre- and postconditions of the transitions and the category of nets, defined there as a pullback, is $\mathsf{Gir}_{L\times L}(\mathbf{Set})$. The games of this note live in $\mathbf{Dial}_{L\times L}(\mathbf{Set})$, and the value is computed componentwise.

6 Conclusions

This note starts from the normal-form reading of dialectica objects, in which witnesses and challenges are the strategies of the two players, and asks how much of the play can be recovered inside $\mathbf{Dial}_L(\mathbf{Set})$. One alternation is already present in the morphisms, since their backward maps see the witness, and the functor from $\mathbf{Dial}(\mathbf{Set})$ to $\mathsf{Gir}(\mathbf{Set})$ used in [9] induces the reader comonad, in which every challenge may depend on the commitment. Further alternation comes from the substitution tensor: one application of it is one round of play, its iterates are alternation of bounded depth, and the interchange map schedules independent dialogues into common rounds.

The results come from doing for witnesses what had already been done for challenges. In the modality of $\mathsf{Gir}(\mathbf{Set})$ and in the reader comonad $(U, X^U, \hat{\alpha})$ the challenges are allowed to depend on the witness; in the substitution tensor and in the object $HA = (U^{X^*}, X^*, \gamma)$ the witnesses are allowed to depend on the challenges, with the exponent moved to the other side. On the side of the witnesses, HA is the cofree $\triangleleft$-comonoid on A (Proposition 2), so the passage from normal to extensive form is made inside the category, and the modality $\dagger A$ of the linear logic model of state [4] maps into it by the universal property. Since the proof uses the witness in the backward maps, it holds in $\mathbf{Dial}_L(\mathbf{Set})$ and not in $\mathsf{Gir}_L(\mathbf{Set})$. The two sides differ in what they do to the value of a game, which is a lax monoidal functor to the lineale. An exponent on the side of the witnesses can raise the value, and only over lineales whose multiplication does not preserve meets (Proposition 3). An exponent on the side of the challenges never changes the value but closes the gap above it, so that over a completely distributive lineale every game in which the opponent replies to the commitment has a value (Proposition 4). Saddle points, the equilibria of games with a single payoff, are closed under both tensors, and the values of two games with saddle points multiply over any lineale. A game with a payoff for each player is an object over the product lineale, which is the setting of the Petri nets of Elena Di Lavore,

Wilmer Leal and de Paiva [11].

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